Programming Project 1-2 – Multi-Threaded Echo Server [**40 pts**]

COMP7212 Operating/Distributed Systems

Due: 10/16/2026

1. Project Objective

In this project, based on the course materials on multi-threading for distributed systems, students will build a server that can handle multiple client connections concurrently, echoing back any messages it receives from them.

2. Project Description

In this project, students are required to develop a Multi-Threaded Echo Server. The primary task is to create a server that can listen for incoming TCP connections and handle each connection in a separate thread. For each connected client, the server should receive messages and send back the same messages as an echo response. Students must implement using the inter-process communication mechanism of their preferred programming language for network communication and the threading library to manage multiple client connections concurrently.

The project involves setting up the server to continuously accept connections, creating a new thread for each client that handles the communication, and ensuring that the server can manage multiple simultaneous connections without crashing or losing data integrity. The final requirement is to test the server with multiple clients to ensure it functions correctly under different scenarios and document the setup and usage instructions clearly.

3. Testing Your Server

To test the multi-threaded server, you can use a simple TCP client that connects to the server, sends messages, and receives the echoed response. To get full marks, you should be able to open multiple terminal windows and run different instances of the client to simulate multiple clients.

4. What to Submit

- page document explaining how to execute your project
- Source code for your server and client
- (Optional) Any other files that are needed to run your project.